

Al-Aqsa University

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جامعة الأقصى
كلية الحاسبات وتكنولوجيا المعلومات
قسم الهندسة والشبكات

Operating Systems for Intelligent Systems (ENGS2306)

Lecture 3: Linux File System and Permissions

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File Systems in General

- **What is a File System?:**
 - A method for organizing, storing, and retrieving data on storage devices (e.g., HDD, SSD).
 - Defines how files are named, stored, and accessed.
 - **Purpose:**
 - Manages data integrity, access speed, and storage efficiency.
 - Provides a logical structure for users and applications.
 - **Common Types:**
 - **FAT32:** Used in USB drives; limited file size (4GB max).
 - **NTFS:** Windows default; supports large files and encryption.
 - **ext4:** Linux default; robust for servers and embedded systems.
 - **Btrfs:** Advanced Linux file system with snapshots.
 - **Relevance to Intelligent Systems:**
 - File systems optimize data access in IoT devices, Android, and embedded systems.
 - Choice of file system impacts performance (e.g., ext4 for Linux-based IoT).
 - **Example:** Android uses ext4 for internal storage, ensuring fast app access.
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Linux File System Details

- **Hierarchy:**
 - Starts at root (/); all directories branch from here.
 - Key directories:
 - /home: User data (e.g., /home/student).
 - /etc: Configuration files (e.g., /etc/passwd).
 - /var: Logs and variable data (e.g., /var/log/syslog).
 - /bin: Essential binaries (e.g., ls, cat).
 - /dev: Device files (e.g., /dev/sda).
 - /proc: Virtual file system for process info (e.g., /proc/cpuinfo).
 - /tmp: Temporary files, cleared on reboot.
- **File Types:**

- Regular files (e.g., file.txt).
 - Directories (e.g., /home).
 - Symbolic links (e.g., shortcuts to files).
 - Device files (e.g., /dev/null).
 - **Relevance:**
 - /proc provides runtime data for IoT monitoring.
 - /dev manages hardware in embedded systems.
 - **Example:** cat /proc/cpuinfo shows CPU details for optimization.
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Linux User Types

- **User Categories:**
 - **Root User:**
 - Superuser with full access (UID 0).
 - Used for system administration (e.g., sudo).
 - **Regular Users:**
 - Created for individuals (e.g., student, UID ≥ 1000).
 - Limited permissions for security.
 - **System Users:**
 - For services (e.g., www-data, UID 1–999).
 - Restricted, no login access.
 - **Viewing Users:**
 - id: Shows UID, GID, groups (e.g., id student).
 - cat /etc/passwd: Lists users and properties.
 - **Relevance:**
 - User types ensure secure access in multi-user intelligent systems.
 - Root configures IoT; regular users manage app data.
 - **Example:** sudo chown root /etc/config restricts to root.
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Exploring the File System

- **Commands for Navigation:**
 - `ls -l`: Lists files with details (type, permissions, owner).
 - `cd /dir`: Navigates to directory (e.g., `cd /var`).
 - `tree`: Displays structure (install: `sudo apt install tree`).
 - **Examples:**
 - `ls -l /dev`: Shows device files (e.g., `sda`, `null`).
 - `cd /proc; ls`: Lists process info (e.g., `cpuinfo`).
 - `tree -L 1 /etc`: Shows `/etc` structure.
 - **Why It Matters:**
 - Navigation of `/dev`, `/proc` is key for hardware/process management.
 - Prepares for Week 13's log analysis mini-project.
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File and Directory Permissions

- **What are Permissions?:**
 - Control read (r), write (w), execute (x) for files/directories.
 - Assigned to owner, group, others.
 - **Viewing Permissions:**
 - `ls -l`: Shows permissions (e.g., `drwxr-xr-x`).
 - `d`: Directory; `-`: File.
 - `rw`: Permissions for owner, group, others.
 - Example: `-rw-r--r--` student users file.txt (owner: rw, others: r).
 - **Relevance:**
 - Permissions secure data in intelligent systems (e.g., IoT configs).
 - Misconfigurations cause vulnerabilities.
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Managing Permissions with `chmod`

- **Purpose:** Change permissions (`chmod` = change mode).
- **Syntax:**

- Symbolic: `chmod u+rwx file` (add read/write/execute for user).
 - Numeric: `chmod 755 file` (owner: rwx, group/others: rx).
 - r=4, w=2, x=1 (e.g., `rwx=7`, `rw-=6`).
 - **Examples:**
 - `chmod u+w file.txt`: Add write for owner.
 - `chmod 644 file.txt`: Owner (rw), group/others (r).
 - `chmod -R 755 my_dir`: Recursive application.
 - **Use Case:** Secure /etc files in IoT devices.
 - **Tip:** Use `man chmod` for syntax.
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Managing Ownership with chown

- **Purpose:** Change owner/group (`chown` = change owner).
 - **Syntax:**
 - `chown user file`: Change owner.
 - `chown user:group file`: Change owner and group.
 - `chown -R user:group dir`: Recursive.
 - **Examples:**
 - `chown student file.txt`: Set owner.
 - `chown student:users /home/student/data`: Set owner/group.
 - `ls -l`: Verify.
 - **Use Case:** Assign ownership in multi-user systems.
 - **Tip:** Use `sudo` for system files.
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Practical File System and Permissions Scenario

- **Task:** Analyze and manage file system and permissions.
 1. Open terminal (`Ctrl+Alt+T`).
 2. Run `ls -l /home` to view user directories/permissions.
 3. Navigate: `cd /var/log`; `ls -l` to inspect logs.
 4. Explore: `ls -l /dev` for device files.

5. Create: `mkdir ~/my_files`.
 6. Create: `touch ~/my_files/config.txt`.
 7. Check: `ls -l ~/my_files`.
 8. Set permissions: `chmod 640 ~/my_files/config.txt` (owner: rw, group: r).
 9. Set ownership: `chown student:users ~/my_files/config.txt`.
 10. Verify: `ls -l ~/my_files; id`.
- **Goal:** Practice navigation, permission, user management.
 - **Record:** Save outputs for Assignment 1.
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Idea Behind the Assignment

- **Learning Objectives:**
 - **Understand File Systems:** Compare general file systems (ext4, NTFS) and Linux hierarchy (`/dev`, `/proc`) for data organization.
 - **Master User Context:** Identify user types (root, regular, system) for secure access in intelligent systems.
 - **Apply Permissions:** Use `chmod`, `chown` to control access, simulating IoT/Android security.
 - **Develop Analytical Skills:** Analyze structures, permissions, users for system security.
 - **Why It Matters:**
 - File systems/users are key for IoT logs, Android data.
 - Prepares for Week 13's mini-project.
 - **Insight:** Simulates securing multi-user systems.
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Week 2 Assignment

- **Assignment 1:** Analyze File Systems, Users, Permissions
 - **Tasks:**
 1. **Explore File System:**
 - Run `ls -l /home`, `ls -l /etc`, `ls -l /var/log`, `ls -l /dev`, `ls -l /proc`.
 - Use `tree -L 1 /etc` or `ls -l /etc` for structure.
 - Identify 3 `/etc` files (e.g., `hostname`, `passwd`, `fstab`) and purposes.

- Run `cat /proc/cpuinfo`; note CPU detail.

2. Analyze Users:

- Run `id` for UID, GID, groups.
- Run `cat /etc/passwd | grep -E '^(root|[^:]*:x:[0-9]{4,})'` for root/regular users.
- Identify system user (e.g., `www-data`) and purpose.

3. Create/Manage Structure:

- Create: `mkdir ~/test_dir; touch ~/test_dir/file1.txt ~/test_dir/file2.txt ~/test_dir/config.txt`.
- Permissions: `chmod 640 ~/test_dir/file1.txt, chmod 755 ~/test_dir, chmod 600 ~/test_dir/config.txt`.
- Ownership: `chown student:users ~/test_dir/file2.txt`.

4. Analyze Permissions:

- Run `ls -l ~/test_dir` to verify.
- Test `config.txt` access as another user (if possible).
- Explain `config.txt` restriction (600).

5. Submit:

- Compile outputs (screenshots/text, e.g., `ls -l > output.txt`).
- Write report (1–2 paragraphs) on permission setting (e.g., 640) and user type.
- Submit via Moodle.

• Tips:

1. Use `man chmod`, `man chown`, `man passwd`.
2. Save outputs: `ls -l ~/test_dir > output.txt`.
3. Screenshots with `PrtSc`.

- **Deadline:** End of Week 2.
- **Weight:** 12% of practical grade.

Next Steps

• Complete Tasks:

- Practice `ls -l`, `cd`, `chmod`, `chown`, `tree`, `id`.

- Experiment with permissions (e.g., `chmod 600 file`).
 - Save VMware snapshots.
 - **Review:**
 - Rewatch Lecture 3 on YouTube/Moodle.
 - Use man pages, Ubuntu forums.
 - **Prepare for Week 3:** Users and Groups Management (creating users, groups, managing access).
 - **Support:** Contact Eng. Mohammed E. Alhams via Moodle.
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Additional Resources

- **Ubuntu Documentation:** <https://help.ubuntu.com>
- **Linux File System Guide:** https://tldp.org/LDP/intro-linux/html/sect_03_01.html
- **Permissions Tutorial:** <https://www.linux.com/training-tutorials/understanding-linux-file-permissions/>
- **Linux Users Guide:** https://tldp.org/LDP/intro-linux/html/sect_03_02.html
- **Moodle:** Lecture videos, ISOs, submissions.
- **Tip:** Download during stable internet hours in Gaza.